Ex: 12
Date: 8/6/21

12. End - End communication at Transport Layer.

AIM:
implement echo client/server using TCP/UDP sockets.

program: TCP.

```
import socket
import threading.

def - start - server():
  server = socket.socket(socket.AF - INET, socket. SOCK_STREAM
server.bind(('127.0.0.1', 5000))
server.listen(1)
print("[server] Listeing on 127.0.0.1: 5000")
conn, addr = server.accept()
print(f"[server] conneted by {addr}")

while True:
   data = conn.recv(1024).decode()
   if not data or data.lower() == 'quit':
print("[server] connetion closed by client.")
      break;
print(f"[server] Received: {data}")
conn.send(f"server received: {data}".encode())
  conn.close()
server.close()
```

path.

client side:

```
def start_client()
    client = socket.socket(socket.AF_INET, socket.SOCK_STR
    client.connect(('127.0.0.1' 5000))
    print("[CLIENT] connected to server")

    while True:
        msg = input("Enter message (or 'quit' to exit): ")
        client.send(msg.encode())
        if msg.lower() == 'quit':
            break;
        response = client.recv(1024).decode()
        print(f"[server response] {response}")
    client.close()
```

—x—x— main —x—x—

```
if __name__ == "__main__":
    server_thread = threading.Thread(target = start
    daemon = True)
    server_thread.start()
    start_client()
```

M)

output:

[server] listening on 127.0.0.1:5000 [client] connected to server.

[server] connected by ('127.0.0.1', 53458) Enter message (or 'quit' to exit): hellow.

[server] Received: hellow

[Server response] Server received: hellow.

Enter message (or quit to exit): quit.

[Server] connection closed by client.

-server,

Result:

Implementation of TCP ~~UDP~~ has completed & executed successfully.

Ex: No: 12

Date: 8/10/25 Implementation chat client server using UDP sockets.

AIM:

Implementation 'chat client server using UDP sockets

udp code:

```
import socket
import threading

# --- server ---

def udp_server():
    s = socket.socket(socket.AF_INET, socket.SOCK_DGR
    s.bind(('127.0.0.1', 12000))

print("[server] Running on 127.0.0.1:12000")

while True:

    data, addr = s.recvfrom(1024)

    msg = data.decode()

if msg.lower() == 'quit':

    print(f"[server] client {addr} ended the
connection.")

    break.
```

```
print [f" [server] Received from {addr} : {msg} ")
s . sendto ( data , addr) # · Echo back.

  S. close ()
  print ( " [server] Stopped.")

# — x —  client — x —

def udp_client ():
  c = socket. socket ( socket . AF_INET, socket. sock_DGRA

server = ('127. 0. 0. 1' , 12000)

while True:

  msg = input ("Enter message (or 'quit' to stop): ")
  c. sendto ( msg . encode (), server )

if msg . lower () == 'quit':

  print ("[Client] closing connection") break.

data, _ = c. recvfrom (1024)

print ["Echo from server:", data. decode ()]

c. Close()

print ( " [Client] stopped.")
```

— x — main — x —

```
if __name__ == "__main__":
    threading.Thread(target = udp_server, daemon=True).
    start(). udp_client().
```

output:

```
Enter menge ( or 'quit' to stop ): [server] Running on 127.
[server] Received from ('127.0.0.1', 49670): hi
echo from server: hi
Enter menge ( or 'quit' to stop ): hello
[server] Received from ('127.0.0.1', 49670): hello
echo from server: hello
Intn menge ( or 'quit' to stop): quit
```

Result:

Implementation of UDP has executed successfully.